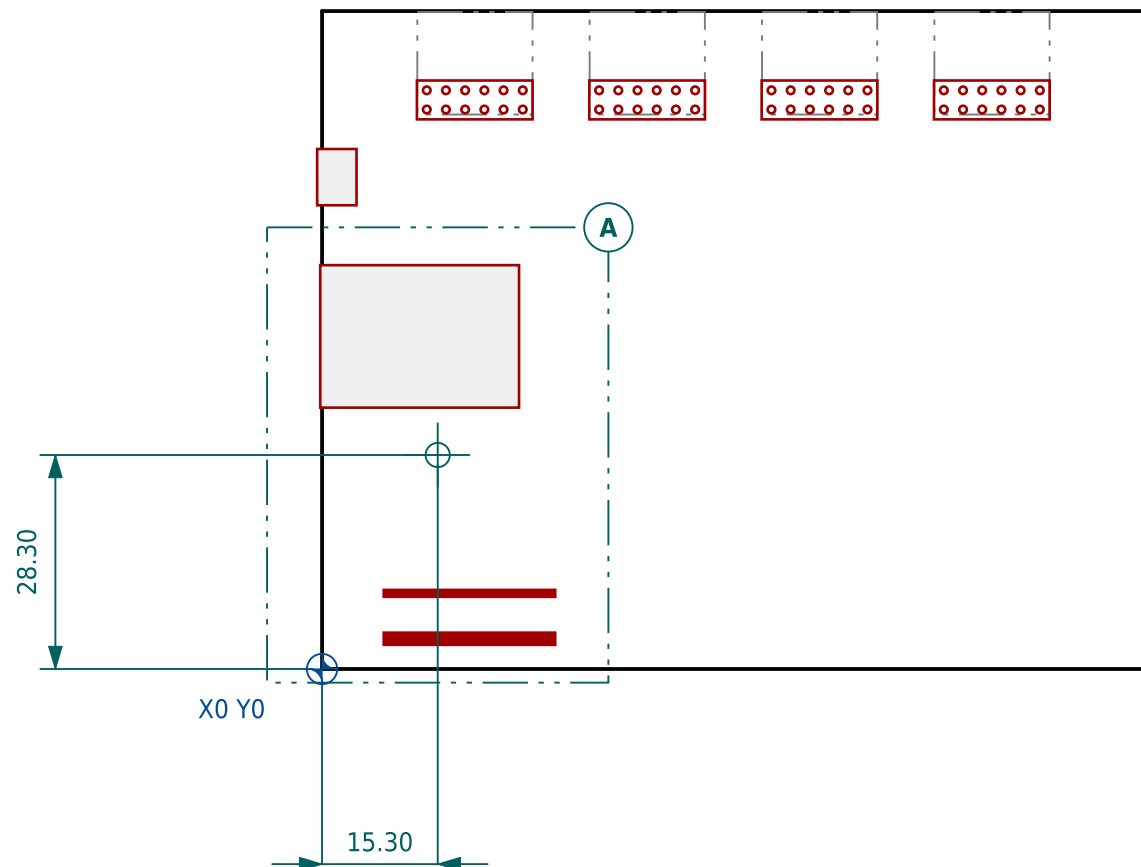




FRAMES, AND Z ABOVE THE FRAME PLANE, mm

	FRAME	RECTANGLE mm	LONG	AXIS X	AXIS Y	Z 65	Z 120
A	LD0-LD7 + RJ45	45.15 x 60.21	Y	15.30	28.30	59.7	22.6

LENSES

LENS	H deg	V deg	FOCUS	NEAR mm
65 deg (stock lens)	53.50	41.41	Fixed	1000
120 deg (wide lens)	120.00	90.00	Fixed	not published
Autofocus	54.00	44.00	Auto	not published

LEGEND

	Subject outline
	LED, display or connector
	Adjacent part or connector body
	Frame A, the rectangle the picture must cover
	Dimension, extension and leader

SOURCES (continued)

- Raspberry Pi camera documentation:
<https://web.archive.org/web/20241230011811/https://www.raspberrypi.com/documentation/accessories/camera.html> - Camera Module 1 column: the "OmniVision OV5647", its "2592 x 1944 pixels" at "1.4 um x 1.4 um", "3.76 x 2.74 mm", "3.60 mm +/- 0.01", "53.50 +/- 0.13 degrees", "41.41 +/- 0.11 degrees", focus "Fixed", "Approx 1 m to infinity".
- Arducam 5MP OV5647 documentation:
<https://docs.arducam.com/Raspberry-Pi-Camera/Native-camera/5MP-OV5647/> - Product catalogue: B0033 "Stock Lens 54 (H) x 41 (V) Fixed Focus", B006604 "120 (H) x 90 (V)", B0176 "54(H)x44 (V) Auto Focus".
- Arducam motorized focus camera:
<https://docs.arducam.com/Raspberry-Pi-Camera/Motorized-Focus-Camera/Motorized-Focus-Camera/> - Quoted: "you can understand it the same as autofocus"; its OV5647 guide adds a voice-coil device tree line and a close-focus range.

CAMERA OVER THE ARTY A7, ETHERNET LEDS INCLUDED

Camera Module OV5647, 65 and 120 degree lenses

DRAWN generated		VERSION v0.0-326-gcbe28df		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-OVER-ETH				REV A
SUBJECT Digilent Arty A7			GENERAL TOLERANCE jack body +/-0.30, pipe exit ASSUMED, Z DERIVED				

NOTES

- The camera looks straight down. X and Y locate the optical axis in the subject's plan frame; Z is the lens height above the plane its target lies in. ASSUMED: Z is to the entrance pupil, which no vendor locates; set it from the lens face.
- A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round; LONG says which way the camera is turned. It is the shape of the picture, not of the optics, so both lenses share it and differ only in Z.
- PLANE, which Z is measured from: A from the subject's own top face. A stand set h mm too low covers only (Z - h) / Z of what is drawn, so the error always loses the edges.
- FOCUS: Raspberry Pi give the depth of field as "Approx 1 m to infinity", so 1000 mm is the nearest this lens focuses. Every height here is nearer than that, so a stock OV5647 cannot focus on any of these boards. Use one whose lens focuses closer.
- AUTOFOCUS: an autofocus OV5647 is a different part -- Arducam's B0176, declared 54 x 44 deg against the stock lens's 53.50 x 41.41, with a motorised lens and a close focus setting. Arducam publish no near limit and no lens height for it. Raspberry Pi's focusable modules, other sensors, are given as "Approx 10 cm to infinity" and "Approx 5 cm to infinity".
- The 120 deg lens's declared pair is not self-consistent: 120 across a 4:3 sensor implies 104.82 down it, not 90. Z is the greater of what the two ask for, so the picture runs OVER the rectangle -- A by 18.00 mm in Y -- which is still covered.

NOTES (continued)

- The "65 deg" name is the stock lens's diagonal, which no vendor prints. DERIVED: $2 \times \text{atan}(\sqrt{3.76^2 + 2.74^2}) / 2 / 3.60 = 65.74$ deg from the image area, 64.42 from the pixel array.
- Quotes here are transliterated to ASCII: x for the multiplication sign, um for micro, infinity for its symbol. The cached pages carry them as printed.
- Frame A, LD0-LD7 + RJ45: The user LEDs together with the Ethernet jack body J9, which is where the link and activity LEDs are.
- ASSUMED: the Ethernet LEDs are on the front face of the RJ45 jack, pointing out of the board edge, and cannot be seen from above at all. This frame covers the jack's BODY footprint, on the assumption that a light pipe adapter brings them to the top somewhere near it. Nothing here is a measurement of such an adapter.
- FPGA-LP-ARTY draws one, and it puts its pipe tips 5.01 mm in FRONT of the jack's front face rather than over the body. That is inside this frame, which reaches further out again, but the frame is still the jack's body and the margin: treat its extent as provisional until an exit is dimensioned onto this sheet.

SOURCES

- Board geometry: fpga/boards.py - Outline, Pmod hosts, connectors and LED rows; see FPGA-ARTY-A7.